Guess The Number

A collaborative Python game with progressive difficulty

How It Works

Guess a randomly generated number before your attempts run out. Three teammates contributed code to create this complete game.

Features:

- Three difficulty levels
- Score tracking
- High score saving
- Hint system

Progressive Difficulty System

Easy

1-20 range, 5 attempts

Medium

1-50 range, 5 attempts

Hard

1-100 range, 5 attempts

Difficulty increases automatically as you win more rounds

Game Mechanics

Correct Guess

+10 points, advance level

Wrong Guess

-1 point, lose attempt

Game Over

Score saved to file

Hint Feedback System

🔥 Very Close

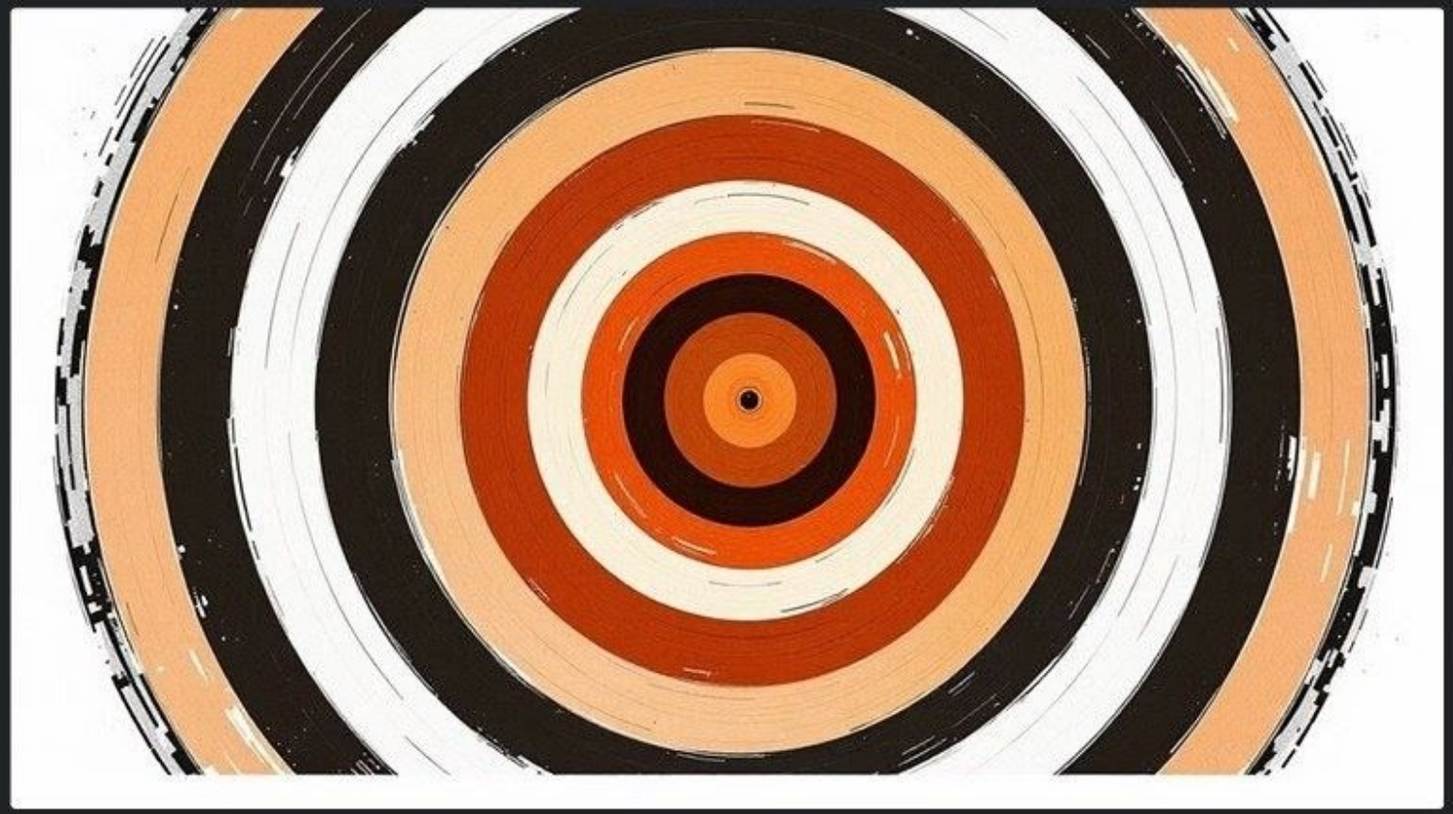
Within 2 numbers

🙂 Close

Within 5 numbers

↗️ High / ↘️ Low

Within 20 numbers



Score & High Score

+10

Correct Answer

Points awarded per win

-1

Wrong Guess

Points deducted per attempt

∞

High Score

Stored in external file

High score persists between game sessions using file I/O

Team Collaboration

01

JIYA

File handling & game loop

02

KHUSBOO

Difficulty progression

03

KANISHKA

Guessing logic

04

VANSHIKA

Feedback system

Technical Features

File Persistence

- Read high score at startup
- Write new high score to file
- Automatic save on game end

Input Validation

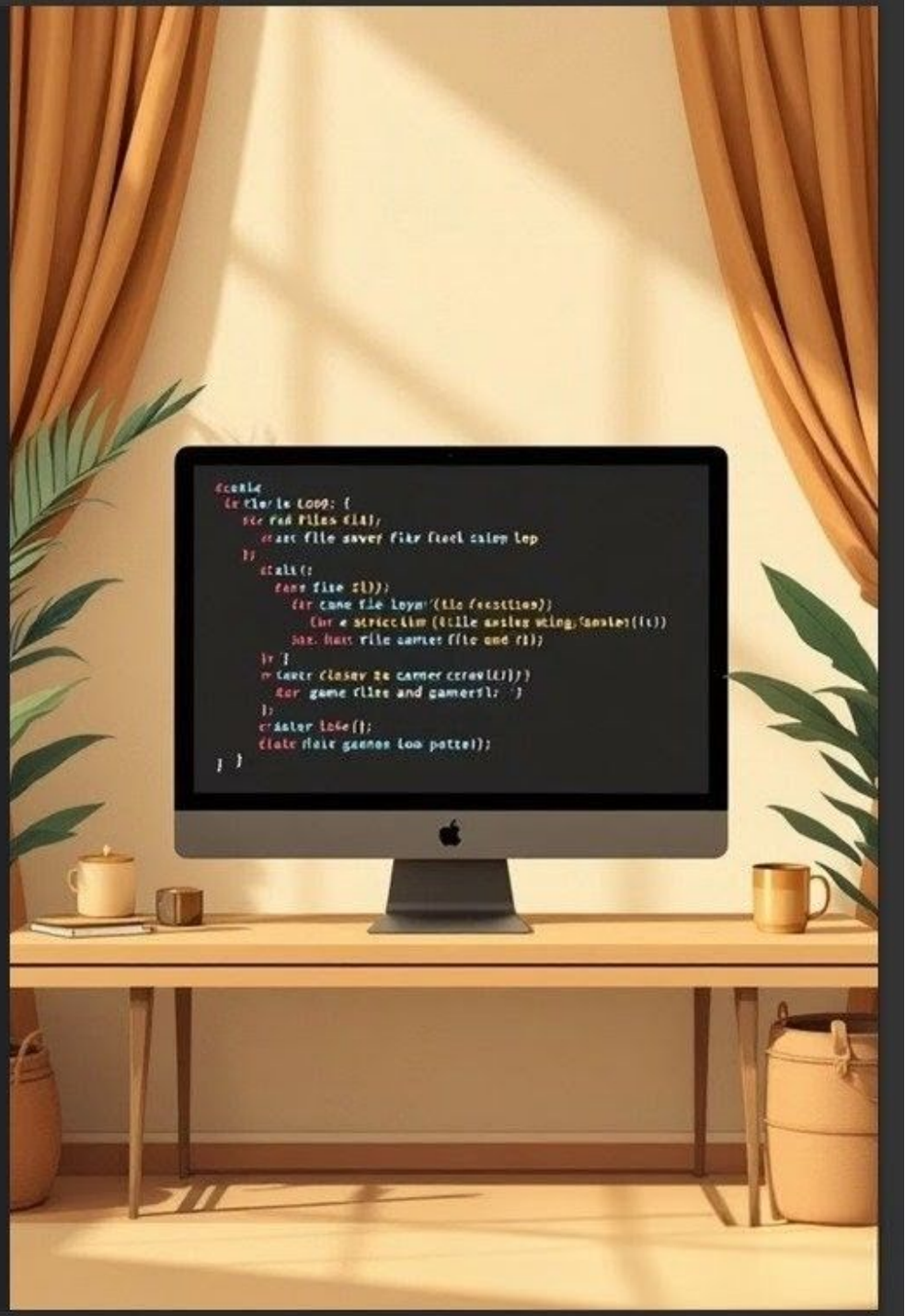
Integer conversion with error handling

Dynamic Variables

- Range changes with wins
- Score updates in real-time
- Attempts tracked per round

Restart System

Play again or exit with final results



Key Learnings

File I/O Operations

Reading and writing data to external files for persistent storage

Loop Control

Nested while loops for game and round management

Conditional Logic

Multiple if-elif chains for game state and difficulty

Variable Scope

Tracking score, wins, attempts across game states

Try It Out!

Download the code and test your guessing skills

Can you beat the high score? 🏆